

5.4 Gravitational fields

This section provides knowledge and understanding of Newton's law of gravitation, planetary motion and gravitational potential and energy.

Newton's law of gravitation can be used to predict the motion of orbiting satellites, planets and even why some objects in our Solar system have very little atmosphere with the opportunity to analyse evidence and look at causal relationships (HSW1, 2, 5, 7).

Geostationary satellites have done much to improve telecommunications around the world. They are expensive; governments and industry have to make difficult decisions when building new ones. Learners have the opportunity to discuss the societal benefits of satellites and the risks they pose when accidents do occur (HSW9, 10).

5.4.1 Point and spherical masses

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
(a) gravitational fields are due to objects having mass	
(b) modelling the mass of a spherical object as a point mass at its centre	
(c) gravitational field lines to map gravitational fields	HSW1
(d) gravitational field strength; $g = \frac{F}{m}$.	
(e) the concept of gravitational fields as being one of a number of forms of field giving rise to a force.	Learners will be expected to link this with section 6.2